COIL TILE / USB INDUCTIVE SENSOR ARRAY

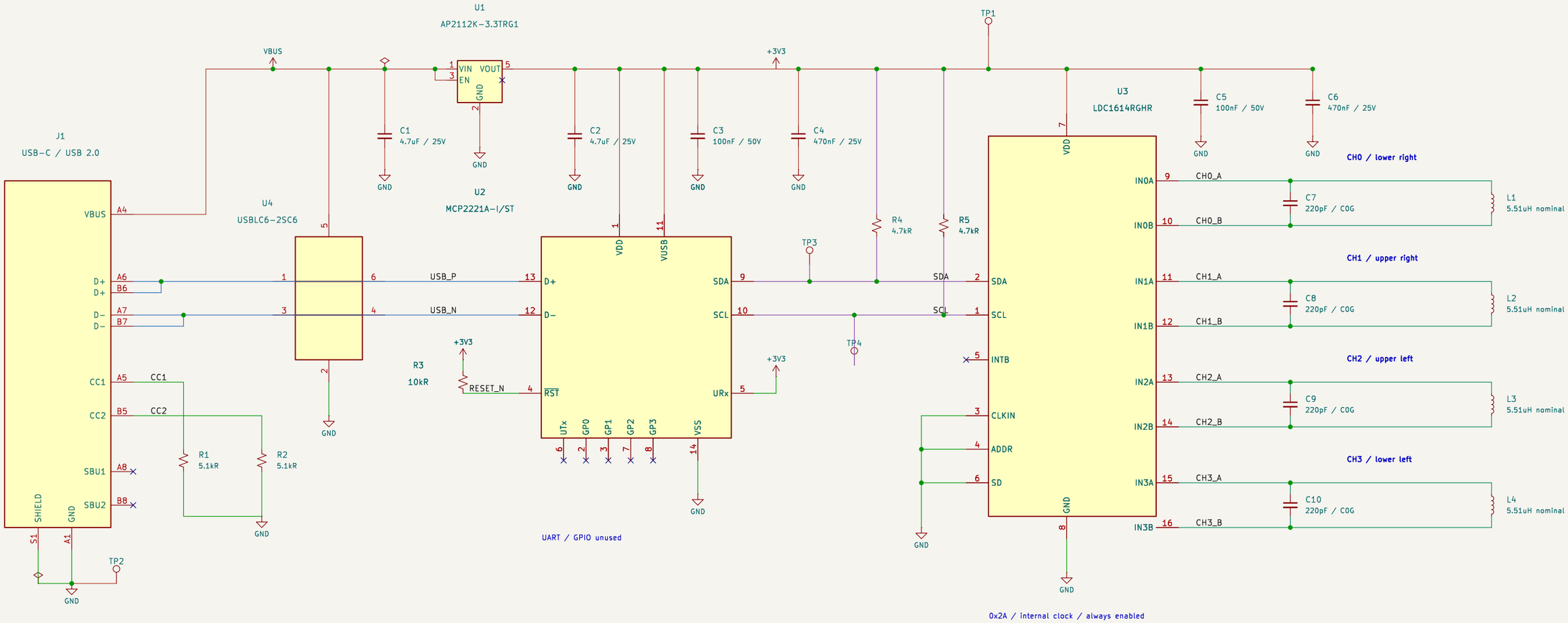
A3 prototype | 2 x 2 printed coils | USB powered | No firmware programming

01 USB + PROTECTION

02 HID / I2C BRIDGE

03 INDUCTANCE CONVERTER

04 LC SENSORS



BUILD + SERVICE

USB-C: 5 V sink; independent 5.1k CC pull-downs.

MCP2221A: VDD and VUSB both supplied from external 3.3 V.

TP1: 3V3 TP2: GND TP3: SDA TP4: SCL

SENSOR NOTES

24 mm / 16 turns / 0.20 mm trace + gap / 220 pF C0G

5.51 uH and ~4.57 MHz nominal; nearby metal changes L and Q.

Unbuilt A3: characterize amplitude, noise and coil coupling.

H1 M3 / NPTH 3.2mm

H2 M3 / NPTH 3.2mm

H3 M3 / NPTH 3.2mm

H4 M3 / NPTH 3.2mm

M3 exposed-substrate edge holes / American Embedded

Same mounting geometry as flyback + GaN examples. Use nylon hardware.

USB HID / Planar Studio

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Title: CoilTile – 2 x 2 inductive sensing prototype

Size: A3	Date: 2026-10-12	**Rev: A3**
KiCad E.D.A. 9.0.9		Id: 1/1